

GSB (Granular Sub■Base)

GSB (Granular Sub■Base) – Construction Material Description & Specifications

Granular Sub■Base (GSB) is an essential structural layer used in road and highway construction. It lies between the prepared subgrade and the Wet Mix Macadam (WMM) or base layer. The GSB layer increases pavement stability, provides a strong foundation, and ensures drainage to protect upper pavement layers.

1. MATERIAL DESCRIPTION

- GSB consists of well■graded granular materials such as crushed stone, natural gravel, crushed slag, or a blend of these.
- It must be free from organic matter, clay lumps, and excessive fines.
- The material should be hard, durable, and non■plastic.

2. GRADATION REQUIREMENTS (As per MoRTH)

Typical GSB grading requirements include:

- 75 mm – 100% passing
- 26.5 mm – 55–75% passing
- 4.75 mm – 10–30% passing
- 0.425 mm – 3–10% passing
- 0.075 mm – 0–5% passing

The gradation may vary slightly based on project specifications.

3. PHYSICAL REQUIREMENTS

- Aggregate Impact Value (AIV): $\leq 40\%$
- Liquid Limit: $\leq 25\%$
- Plasticity Index (PI): $\leq 6\%$
- California Bearing Ratio (CBR at 98% MDD): $\geq 30\%$

4. CONSTRUCTION PROCEDURE

- Level and prepare the subgrade before placing GSB.
- Spread the material in uniform layers (minimum thickness: 150 mm per compacted layer).
- Use mechanical mixing for uniformity.
- Compact using a vibratory roller to achieve at least 98% of maximum Proctor density.
- Conduct field tests such as density test, gradation check, and CBR test for quality control.

5. APPLICATIONS

- Highways and expressways
- Rural and urban roads
- Industrial yards
- Airport pavements

6. ADVANTAGES

- Provides stable load distribution
- Prevents capillary rise of water
- Enhances pavement lifespan
- Improves drainage characteristics

This document summarizes the essential specifications and construction requirements for GSB used in modern pavement engineering.